

Exploratory Data Analysis

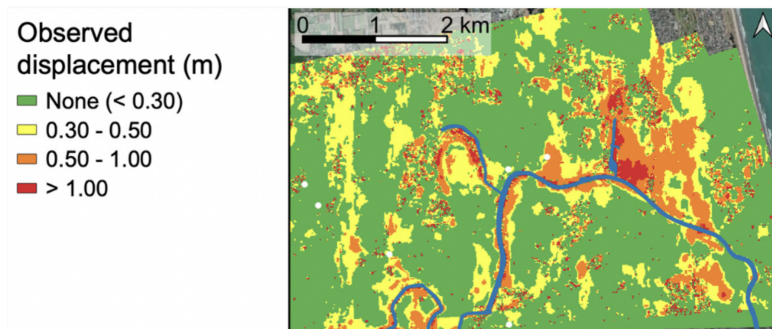
Krishna Kumar & Ellen Rathje

University of Texas at Austin

Overview

- 1 Why EDA Matters
- 2 Understanding Distributions
- 3 Quartiles: Robust Statistics
- 4 Outlier Detection
- 5 Z-Scores: Two Uses
- 6 Correlation and Significance
- 7 Data Quality

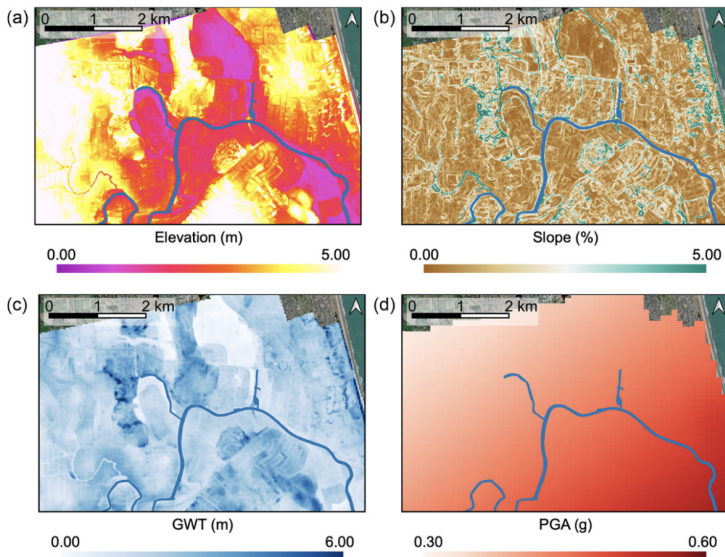
Predicting Lateral Spreading in New Zealand



Goal: Predict lateral spreading displacement during earthquakes

Reference: Durante, M. G., & Rathje, E. M. (2021). An exploration of the use of machine learning to predict lateral spreading. *Earthquake Spectra*, 37(4), 2288-2314.

Input Features for Prediction



Before building models, understand your data!

EDA answers:

- What does the data tell us about the physics?
- Are there quality issues (missing, errors, outliers)?
- Which features predict lateral spreading?
- How should we preprocess?

The Key Question

Dataset: Durante & Rathje (2021)

- 7,291 case histories
- Target: $> 0.3m$ displacement = spreading

Can we distinguish spreading from non-spreading sites?

Approach: Compare distributions by class

- **Overlapping distributions:** Feature not useful
- **Separated distributions:** Feature is predictive

Best visualization: Overlaid histograms

- See both classes simultaneously
- Directly compare shapes and locations
- Immediate intuition about separability

This is more informative than looking at all data together!

Reading Overlaid Histograms

What to look for:

- ➊ **Separation:** How far apart are the peaks?
 - Far apart → strong predictor
 - Overlapping → weak predictor
- ➋ **Direction:** Which class has higher values?
 - GWD: No-spread sites have higher values (deeper water table = safer)
 - PGA: Spread sites have higher values (more shaking = more risk)
- ➌ **Overlap area:** How much do they overlap?
 - Small overlap → easy to classify
 - Large overlap → difficult boundary

Why Quartiles Matter

Problem: Mean and std are sensitive to outliers

Example: Water depths (m): {1.5, 1.8, 2.0, 2.1, 2.3, 15.0}

- Mean = 4.12 m (misleading!)
- Median = 2.05 m (representative!)

Quartiles divide data into four equal parts:

- Q1 (25th percentile): 25% below
- Q2 (Median, 50th percentile): middle value
- Q3 (75th percentile): 75% below
- IQR = Q3 - Q1 (middle 50%)

Box plots visualize the five-number summary

Quartiles for Engineering Decisions: What do these numbers mean?

Example: Ground Water Depth (GWD)

From our data:

- $Q1 = 1.64$ m, Median = 1.98 m, $Q3 = 2.46$ m
- $IQR = 0.82$ m

Engineering interpretation:

- 50% of sites: water table at 1.64-2.46 m depth
- Sites with $GWD < Q1$ (1.64 m): **High risk** zone
- Sites with $GWD > Q3$ (2.46 m): **Lower risk** zone

Action: Quartiles create natural risk categories!

In geotechnical engineering, outliers can be:

- ➊ **Measurement errors:** Remove
 - Negative water table depth
 - Physically impossible values
- ➋ **Extreme events:** Keep!
 - Major earthquakes
 - Unusual but valid site conditions

Never blindly remove outliers!

- They might be your most important data
- Use engineering judgment + statistics

IQR Method (Most Robust)

Rule: Values beyond these bounds are outliers

$$\text{Lower bound} = Q1 - 1.5 \times \text{IQR}$$

$$\text{Upper bound} = Q3 + 1.5 \times \text{IQR}$$

Why $1.5 \times \text{IQR}$?

- John Tukey's empirical rule
- For normal distribution: captures $\sim 99.3\%$ of data
- Balances sensitivity vs over-flagging

What's a good outlier percentage?

- 1-5%: Excellent - typical for clean data
- 5-10%: Good - acceptable natural variability
- $>10\%$: Investigate - may indicate data quality issues

IQR Results for Our Dataset

Outlier analysis using IQR method:

Feature	IQR	Outliers	%	Assessment
GWD	0.82 m	133	1.8%	Excellent
L	0.99 km	8	0.1%	Excellent
Slope	0.95%	542	7.4%	Good
PGA	0.06 g	3	0.0%	Excellent

Analysis:

- ➊ **GWD (1.8%):** Valid extreme water table depths → Keep
- ➋ **L (0.1%):** Very few unusual distances → Keep
- ➌ **Slope (7.4%):** Natural field variability → Keep
- ➍ **PGA (0.0%):** Perfect - earthquake data well-controlled → Keep

Decision: All outliers $< 10\%$ and physically plausible → **Keep all data**

Why? Model needs to learn extreme events (major earthquakes, steep slopes)

What is a Z-Score?

Z-score = how many standard deviations from the mean

$$z = \frac{x - \mu}{\sigma}$$

Interpretation:

- $z = 0$: At the mean
- $z = 1$: One std above mean
- $z = -2$: Two std below mean
- $|z| > 3$: Outlier (only 0.3% of normal data)

68-95-99.7 Rule:

- 68% within $\pm 1\sigma$
- 95% within $\pm 2\sigma$
- 99.7% within $\pm 3\sigma$

IQR vs Z-Score: When to Use Each

Method	Use For	Notes
IQR	Outlier detection	Robust
Z-score	Outlier detection	Assumes normality
Z-score	Standardization	ML preprocessing

Recommendation:

- Detect outliers with IQR
- Standardize features with z-scores for neural networks
- Tree-based models don't need standardization

Correlation Coefficient

Measures linear relationship strength

Properties:

- Range: $r \in [-1, 1]$
- $r = 1$: Perfect positive correlation
- $r = -1$: Perfect negative correlation
- $r = 0$: No linear correlation

Interpretation for liquefaction:

- GWD vs Target: Negative (shallow water = more risk)
- PGA vs Target: Positive (more shaking = more risk)
- L vs Target: Negative (closer to river = more risk)

Warning: Correlation $\neq$ Causation!

Mann-Whitney U Test

Question: Are distributions truly different, or just random chance?

What is U? U counts the “wins” - how often one group beats the other.

The Algorithm:

- 1 **Mix all values** from both groups together
- 2 **Rank them** from smallest to largest (1, 2, 3, ...)
- 3 **Sum ranks** for each group
- 4 **Calculate U:** How often does spreading value j non-spreading value?

Why it's powerful:

- No assumptions about distribution shape
- Robust to outliers (uses ranks, not values)
- Perfect for comparing two groups

Mann-Whitney U Test - Visual Example

Example with GWD (Ground Water Depth):

All sites mixed and ranked by GWD:

Spreading sites: [1.2] [1.5] [1.8] ...
 ↓ ↓ ↓
 rank:1 rank:2 rank:3

Non-spreading sites: [2.5] [3.0] [3.2] ...
 ↓ ↓ ↓
 rank:4 rank:5 rank:6

- Spreading sites get LOWER ranks
- They have shallower water tables!

The p-value answers:

“If groups were actually the same, what’s the chance we’d see this much separation by random luck?”

- $p < 0.001$ → Less than 0.1% chance → **Strong evidence!**
- $p > 0.05$ → More than 5% chance → Could just be randomness

Mann-Whitney Results for Our Data

Statistical significance of each feature:

Feature	p-value	Significant?	Conclusion
GWD	4.76e-15	***	Strong predictor
L	4.90e-46	***	Strongest predictor
Slope	9.83e-02	No	Weak predictor
PGA	1.35e-16	***	Strong predictor

Key Insight:

- **Slope is NOT significantly different** → May not help prediction!
- **GWD, L, PGA** show extremely low p-values → Key predictors

P-value interpretation:

- $p < 0.001$: *** Extremely significant
- $p < 0.01$: ** Very significant
- $p < 0.05$: * Significant
- $p \geq 0.05$: Not significant

Before modeling, verify:

1 Missing values:

- How many? Which features?
- Delete ($< 5\%$) or impute ($> 5\%$)?

2 Physical plausibility:

- Negative values where impossible
- Values outside engineering bounds

3 Duplicates:

- Exact copies (data entry errors)?
- Same site, multiple events (valid)?

4 Class balance:

- 50-50 split? 90-10?
- Severe imbalance needs special handling

Common Mistakes to Avoid

- ❶ **Blindly trusting summary statistics**
 - Always visualize!
- ❷ **Removing outliers without investigation**
 - Might be your most important data
- ❸ **Confusing correlation with causation**
 - Ice cream sales correlate with drowning... both caused by summer!
- ❹ **Ignoring class imbalance**
 - 99% accuracy meaningless for 1% rare class
- ❺ **Skipping domain knowledge check**
 - Does this make engineering sense?

Creating new features from existing data

What features to introduce for liquefaction datasets

① **Interaction features:**

- $\text{GWD} \times \text{PGA}$ (water depth $\times$ shaking intensity)
- $\text{L} \times \text{Slope}$ (distance $\times$ slope)

② **Polynomial features:**

- GWD^2 , PGA^2 (capture non-linear relationships)

③ **Domain-specific features:**

- L/GWD (normalized distance)
- $\text{is_shallow_water} = \text{GWD} < 2 \text{ m}$ (binary threshold)

④ **Binning/Discretization:**

- Convert GWD: “shallow”, “medium”, “deep”

⑤ **Transformations:**

- $\log(\text{L})$, $\sqrt{\text{Slope}}$ (handle skewness)

EDA is the foundation of successful ML

We accomplished:

- Understood data structure and quality
- Identified predictive features
- Compared distributions by class
- Detected and evaluated outliers
- Gained physical insight into liquefaction
- Made informed preprocessing decisions

"No amount of sophisticated modeling can compensate for poor understanding of your data."

Questions?

"All models are wrong, but some are useful."

— George Box

"See the structure first, then calculate."

— Gilbert Strang

Contact:

Dr. Krishna Kumar [krishnak@utexas.edu]

Dr. Ellen Rathje [rathje@mail.utexas.edu]

University of Texas at Austin

Remember:

"Understanding the data is understanding the physics."